Krishna Beniwal

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	6.14
GURU BRAHMANAND SR SEC SCHOOL	Class XII (CBSE)	2024	94.00%
GURU BRAHMANAND SR SEC SCHOOL	Class X (CBSE)	2022	95.40%

SKILLS

- **Programming Languages:** C, C++, Python, JavaScript, HTML, CSS, OCaml
- **Libraries:** discord.py, aiosqlite, asyncio
- **Concepts:** Data Structures and Algorithms, Object-Oriented Programming, Async Programming
- **Databases:** SQL, Database Design
- **Tools & Platforms:** Git, GitHub, LaTeX, Canva

RELEVANT COURSES

CS: Theory of Computation · Algorithms · Compilers · OS · DBMS · Data Structures · Architecture · Software Development · AI
Math: Statistics · Transformation Techniques · Linear Algebra · Probability
EE: Digital Circuits · Basic Electrical Engineering
Other: Digital Fabrication · Language & Logic

PROJECTS

StonksBacktester *Python* | *Event Driven Backtesting* | *NSE* | *Quant Research*

- Built an event driven NSE backtesting engine with Market, Signal, Order, and Fill lifecycles and look ahead bias protection.

- Implemented SMA, EMA, RSI, MACD, Bollinger Bands, Donchian, Z Score, and ROC strategies.

- Modeled Indian market execution costs across major brokers with brokerage, STT, GST, exchange charges, stamp duty, and SEBI fees.

- Added position sizing, risk controls, parameter search, walk forward validation, Monte Carlo testing, audit logs, metrics, and HTML tearsheets.

Crimson *Python* | *discord.py* | *asyncio* | *Regex* | *In memory Caching*

- Event driven Discord bot for real time Dank Memer event detection, routing, and automation.

- Automated mini game solving using regex, anagram decryption, and image based lookup tables.

- Used in memory caching to reduce API calls, latency, and processing overhead.

Cherry *Python* | *discord.py* | *SQLite (aiosqlite)* | *asyncio* | *Regex*

- Asynchronous Discord bot for reminders, activity tracking, and utility features.

- Built a persistent reminder system using SQLite and asyncio for efficient scheduling.

- Implemented Regex parsing and interactive buttons and dropdowns for dynamic settings management.

K Map Solver *C++* | *Boolean Algebra* | *Karnaugh Maps* | *Greedy Algorithm*

- Built a C++ K Map solver for simplifying 2 to 5 variable Boolean functions into minimized SOP and POS expressions.

- Implemented Gray code traversal, don't care support, group detection, and greedy covering for Boolean logic optimization.

K Map Solver *JavaScript | Karnaugh Maps | Quine McCluskey | Verilog*

- Built an interactive K Map solver for 2 to 5 variables with synchronized truth table and K Map inputs.

- Implemented Quine McCluskey minimization for SOP and POS expressions with automatic visual grouping.

- Added Verilog module and testbench generation for logic verification.

EXTRACURRICULAR

National Service Scheme (NSS), IIT Hyderabad

- Completed 89+ hours of community service over two years through the National Service Scheme (NSS).

- Participated in social awareness campaigns, community outreach activities, and volunteer initiatives promoting civic responsibility and public welfare.

Discord Community Staff (Apr 2025 - Present)

- Served as a staff member in a Discord Partnered community with over 181,000 members.

- Moderated discussions, resolved user issues, and helped maintain a safe, organized, and engaging community environment.

- Planned and hosted many community events while collaborating with the moderation team to improve member engagement and overall server operations.